

Where Does SAT Hardness Live? Mean, Tail, Cluster, and Scale Projections of Defect-Field Geometry

Christof Krieg

MAAT Project – Independent Research

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Abstract

Structural defect fields can be compressed in several ways: by their mean level, high-defect tails, connected hotspot clusters, or scale/conditioning terms. This paper asks which of these projections carries useful signal for predicting CDCL hardness in canonical SAT families.

We reuse a fixed SAT-family benchmark with CDCL statistics from Glucose3 and MiniSat22. No new SAT instances are generated and no solver target is retuned. The predeclared target is the CDCL hardness proxy

$$y_{\text{CDCL}} = \log(1 + \bar{c} + 0.10 \bar{d} + 0.01 \bar{p}),$$

where $\bar{c}$, $\bar{d}$, and $\bar{p}$ are mean conflicts, decisions, and propagations across the two solvers.

The main positive result is localization: inside the all-defect model, grouped permutation tests identify tail and scale projections as the dominant signal carriers, with R^2 drops of 0.4044 and 0.3414 respectively. In ordinary cross-validation, the full defect-projection model remains competitive ($R^2 = 0.1996$, Spearman 0.3751) but does not surpass the standard graph baseline ($R^2 = 0.2169$, Spearman 0.4234). The main negative result is transfer: leave-family-out generalization remains weak; even the best overall leave-family-out projection has negative R^2 and only modest Spearman correlation (0.1607).

Thus the benchmark does not solve SAT hardness and does not establish universal transfer. It localizes the current signal: within this benchmark, SAT hardness is better described by high-defect tails and scale-sensitive frustration geometry than by mean scalar compression alone.

1 Purpose

Previous SAT benchmarks showed that multi-scale defect-field features can carry CDCL-hardness signal beyond shuffled nulls, but do not beat a strong standard graph baseline and fail under leave-family-out transfer. The natural next step is a resolution-aware decomposition:

$$F_{\text{multi}} = F_{\text{mean}} + F_{\text{tail}} + F_{\text{cluster}} + F_{\text{scale}}. \quad (1)$$

The present paper asks a narrower empirical question:

Which projection of the defect field actually carries the SAT-hardness signal?

This is intentionally not a new universality claim. It is a localization test. If global scalar scores fail, the next question is not whether the whole framework is immediately wrong, but whether the information has been lost by the chosen projection.

2 Dataset and Target

The experiment uses a fixed table of 395 canonical SAT instances. The families are:

- random 3-SAT;
- planted 3-SAT;
- modular/community 3-SAT;
- random 3-XOR encoded as CNF;
- graph-colouring encodings;
- pigeonhole-principle encodings.

Each instance was solved with PySAT-compatible Glucose3 and MiniSat22 solvers. The target is inherited unchanged from the source benchmark:

$$y_{\text{CDCL}} = \log(1 + \bar{c} + 0.10 \bar{d} + 0.01 \bar{p}) . \quad (2)$$

Here $\bar{c}$ denotes mean conflicts, $\bar{d}$ mean decisions, and $\bar{p}$ mean propagations across the two solvers.

The present paper does not introduce new solver data. It is a downstream decomposition of already generated defect fields and graph features.

3 Projection Families

The defect field is represented through four projection families. The mean projection is

$$F_{\text{mean}} = \langle d_{\text{frust}} \rangle . \quad (3)$$

The tail projection collects high-quantile and intermittency terms:

$$F_{\text{tail}} \sim q_{95}(d_{\text{frust}}) + M_{95} + \text{Gini}(d_{\text{frust}}), \quad (4)$$

where M_{95} is the mass carried by variables in the top five percent of the frustration field.

The cluster projection measures connected high-defect regions:

$$F_{\text{cluster}} \sim \frac{C_{\text{max}}}{N} + \rho_{\text{hotspot}}, \quad (5)$$

where C_{max} is the largest connected hotspot component and ρ_{hotspot} is hotspot edge density.

The scale projection collects conditioning, expansion, and propagation-depth terms:

$$F_{\text{scale}} \sim \log_{10} \kappa(C_{\text{local}}) + d_{\text{expansion}} + d_{\text{propagation}} . \quad (6)$$

The full defect-projection model uses all four families plus the multi-scale score F_{multi} . Controls include density-only, macro-scalar, and standard graph baselines.

4 Evaluation Protocol

The evaluation is deliberately simple:

- five-fold shuffled cross-validation with a fixed random seed;
- ridge regression and random-forest regression;
- leave-family-out testing across canonical SAT families;
- shuffled all-projection nulls;
- grouped permutation tests for mean, tail, cluster, and scale projections.

The main metrics are R^2 , RMSE, and Spearman correlation. Leave-family-out is treated as the strict stress test because it asks whether one SAT-family geometry extrapolates to another.

5 Results

5.1 Cross-Validation

Table 1: Random-forest cross-validation results.

Feature set	R^2	RMSE	Spearman
Standard graph	0.2169	0.8663	0.4234
All defect projections	0.1996	0.8758	0.3751
Macro scalar	0.1898	0.8811	0.4193
Mean + tail	0.1728	0.8904	0.3254
Tail + scale	0.1725	0.8905	0.3793
Scale projection	0.1381	0.9089	0.3304
Tail projection	0.0937	0.9320	0.2982
Cluster projection	0.0657	0.9462	0.1686
Mean projection	0.0575	0.9504	0.2944
Density-only	-0.0434	1.0000	0.2843
Shuffled all-projection null	-0.0666	1.0110	-0.0134

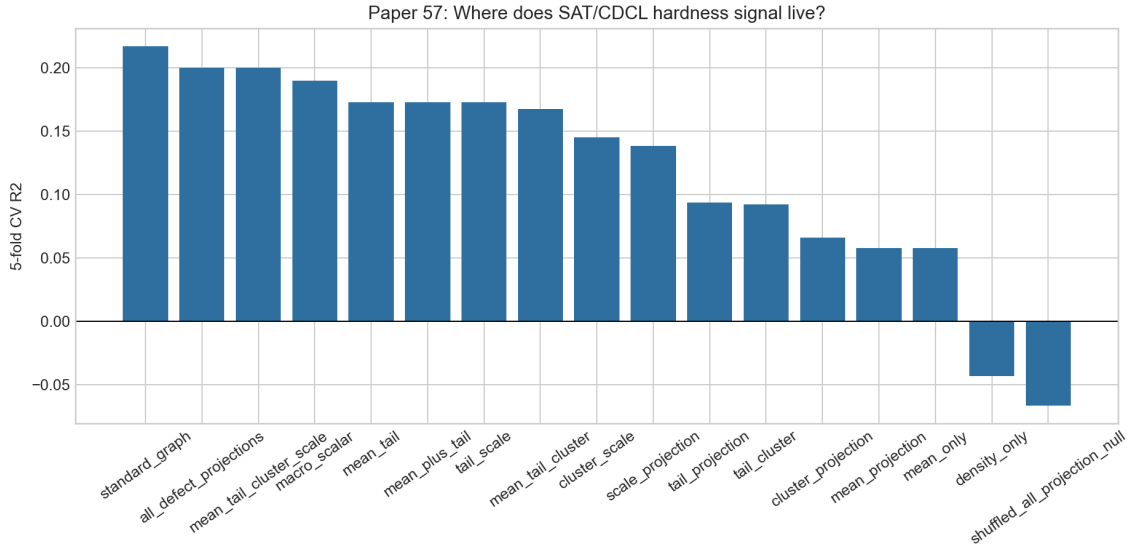


Figure 1: Five-fold cross-validation by projection family. The standard graph baseline remains best, but all defect projections and the macro scalar retain positive signal and strongly beat shuffled-projection nulls.

5.2 Where Is the Defect-Field Signal?

Grouped permutation importance inside the all-defect model gives:

Table 2: Grouped permutation importance inside the all-defect projection model.

Projection	Mean R^2 drop	Std. dev.
Tail	0.4044	0.0149
Scale	0.3414	0.0229
Mean	0.1567	0.0084
Cluster	0.0229	0.0019

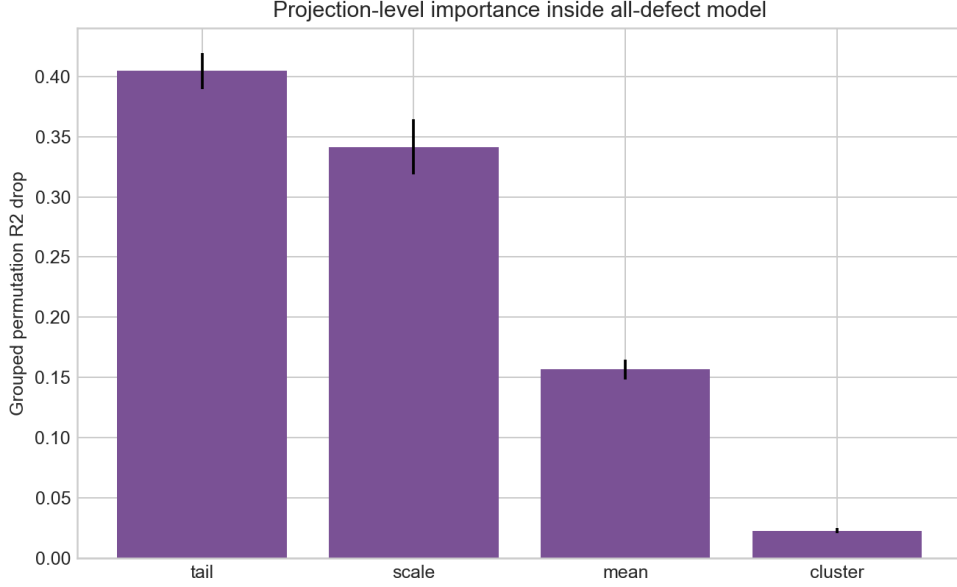


Figure 2: Projection-level grouped permutation importance. Tail and scale projections carry most of the defect-field signal in this benchmark.

This is the clearest result of the paper. Mean compression is not sufficient, and cluster information is weak in this particular implementation. The strongest defect-field information is carried by high-defect tails and scale/conditioning terms.

5.3 Single Features

The strongest individual correlations include macro robustness and density features:

Table 3: Top single-feature Spearman correlations with CDCL hardness.

Feature	Spearman
F_{struct} scalar	−0.3563
R_{rob}	0.3539
Clause density α	0.3459
R_{resp}	0.3331
Mean variable graph degree	0.2972
Frustration q_{75}	−0.2908
Literal imbalance	−0.2897
B	0.2876
Frustration mean	−0.2830
S	0.2698

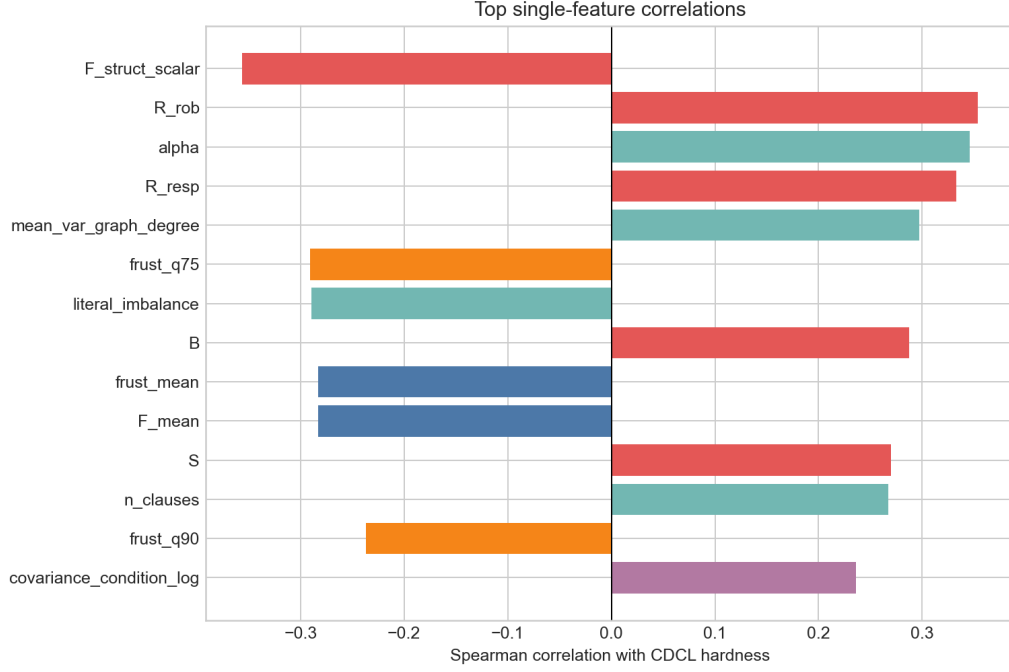


Figure 3: Top single-feature Spearman correlations. No individual feature solves the task; the useful signal is distributed across macro, graph, mean, and tail information.

5.4 Leave-Family-Out Stress Test

Leave-family-out remains the decisive negative result:

Table 4: Overall leave-family-out results.

Feature set	R^2	RMSE	Spearman
Tail projection	-0.1290	1.0402	0.1607
Tail + cluster	-0.1322	1.0416	0.1457
Macro scalar	-0.3128	1.1217	0.1243
Mean + tail	-0.2016	1.0731	0.0437
Density-only	-0.5271	1.2097	0.0019
Standard graph	-2.4937	1.8298	-0.0118
All defect projections	-1.0123	1.3887	-0.0995
Scale projection	-2.4092	1.8075	-0.1930

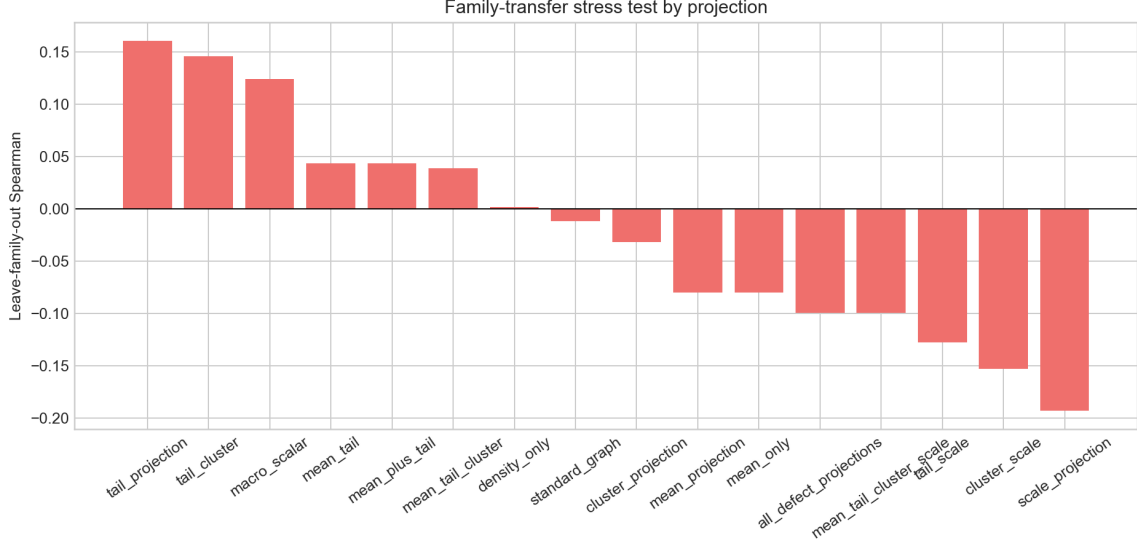


Figure 4: Leave-family-out Spearman correlations. Tail features are the most portable projection in this benchmark, but the overall transfer result remains weak.

6 Interpretation

The result is not that SAT hardness has been explained. It is also not that MAAT defect fields beat standard SAT graph features. The standard graph baseline remains slightly stronger in ordinary cross-validation.

The useful conclusion is more specific:

SAT/CDCL hardness signal is concentrated more strongly in tail and scale projections than in mean defect compression.

This supports the interpretation that global scalar scores are macro-projections of richer defect-field geometry. For smooth systems, a macro-projection may retain enough information. For SAT, the relevant structure appears to involve high-defect variables, intermittent frustration, local conditioning, expansion failure, and propagation geometry.

The weak leave-family-out result is equally important. It shows that the projection language is not yet family-invariant. The model can learn within a mixed canonical SAT population, but extrapolation to unseen SAT families remains unresolved. Future defect-field designs should therefore prioritize tail- and scale-sensitive terms rather than further global compression.

7 Limitations

- The experiment reuses generated SAT families rather than SAT Competition or industrial CNF instances.
- The CDCL target is a transparent proxy built from conflicts, decisions, and propagations, not a complete solver-performance theory.
- The defect-field definitions remain heuristic and were not derived from proof complexity.
- Grouped permutation importance is computed inside an all-defect model and should be read as diagnostic, not causal.
- Leave-family-out transfer remains weak; this limits any universality claim.

8 Reproducibility

The experiment is stored in:

```
experiments/sat_projection_decomposition_paper57/
```

Run:

```
cd experiments/sat_projection_decomposition_paper57
python3 sat_projection_decomposition_paper57.py
```

The script reads:

```
../sat_cdcl_external_paper55/outputs_paper55/
paper55_sat_cdcl_instances.csv
```

and writes CSV/JSON outputs and figures to:

```
experiments/sat_projection_decomposition_paper57/outputs_paper57/
```

No external CNF dataset is redistributed. All inputs are derived reproducibility artifacts generated by Paper 55.

9 Conclusion

This paper answers a concrete projection question in one domain. In the present SAT/CDCL benchmark, hardness signal does not live primarily in a single global scalar score. It is distributed across graph statistics, macro robustness, and especially tail and scale projections of the defect field.

The strongest positive result is localization: tail and scale projections carry most of the internal defect-field signal. The strongest negative result is transfer: leave-family-out generalization remains weak.

Thus the correct next step is not to claim SAT hardness is solved, but to build better domain-stable defect fields for propagation tails, local frustration clusters, and scale-dependent proof-search geometry.

References

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